

more than 20 netlist formats.

gsymcheck is the default syntax checker of the program.

Symbol attributes can be edited on gattrib spreadsheet program.

Libgeda is the gEDA library collection for gschem, gnetlist and gsymcheck.

gsch2pcb suite is responsible for forwarding annotations from the schematic to the layout tool.

PCB. This is the main PCB layout creation program of gEDA suite. It is suitable across all four major platforms (Unix, Linux, Windows and Mac). The tool comprises a gamut of features, including schematic/netlist import, check design rules, auto-router and trace optimiser to reduce layout time.

For board fabrication and assembly operations, the tool can generate a variety of outputs like RS-247X (Gerber) data, NC drill and Centroid data. It can also be customised as required.

gerbv. gerbv is the gerber file viewer of gEDA. gerber is the standard data format for PCB design operations.

NgSPICE. This is a circuit simulator for digital mixed-signal circuits. The created circuit schematic can be simulated on this tool to check the functionalities. The tool is compact, and easy to understand and use. The interface front-end is based on gspiceui setup.

GTKWave and gwave. Along with circuit simulation, gEDA also brings a digital waveform viewer tool, GTKWave, and an analogue waveform viewer, gwave. GTKWave is compatible with Unix, Linux and MacOS. Supported input file types in-

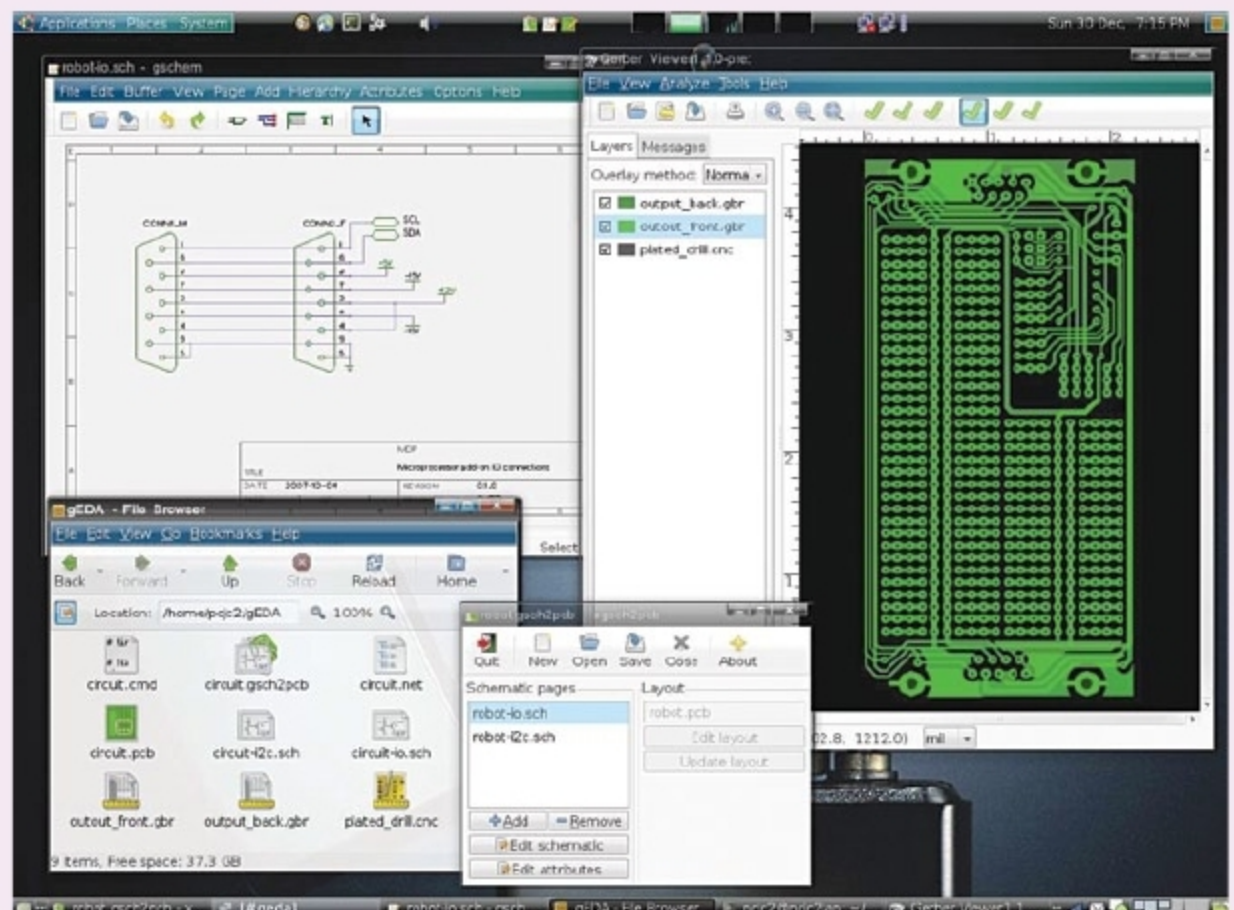


Fig. 3: gerbv running along gschem

clude LXT, LXT2, VT, FST and GHW. Standard Verilog VCD and EVCD files can also be read.

Icarus Verilog. This is a compilation and simulation tool for source codes written in Verilog data. Its main target platform is Linux, but it is compatible with other similar operating systems as well.

Operations and latest features

The main flow of circuit creation on gEDA starts from schematic production on gschem. A netlist is created from the schematic, and attributes are passed on to the PCB (through translation by gsch2pcb). After necessary changes and customisations, the digital circuit can be evaluated on the simulation suites.

gschem suite features a large number of functions and utilities

for various operations, like symbol checkers and symbol customisations, format conversion, attribute editing and so on. Schematics also support hierarchical design.

The latest version of the PCB layout tool has seen many improvements in terms of bug fixes like UI upgrades and dependency modifications. All other tools are provided as per their latest updated iterations.

gEDA has a strong community presence, drawing support from various open source projects. Documentation is available on www.geda-project.org. It is, overall, a complete suite that is highly scalable due to all the assembled open source tools. gEDA suite can be a great companion for electronics enthusiasts of all kind—students, hobbyists or professionals. **EFY**

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